

A. **Infinite Primes.** It is natural to wonder, is there a largest prime?

Theorem. There are infinitely many primes.

This proof is due to Euclid.

Proof. Assume towards a contradiction there are only finitely many primes

Consider the integer

$N =$

Select a prime factor:

Then

and thus:

But this is impossible, so we have arrived at a contradiction.

□

B. **Conjectures.** The distribution of primes does not follow a simple pattern.

Many **conjectures** have been made about the nature of primes.

Twin primes are pairs of primes that are:

For example: 11 and 13, 17 and 19, or 101 and 103

Twin Primes Conjecture. There exist

In the 1700s, Prussian mathematician Christian Goldbach proposed

Goldbach Conjecture. Every **even** integer greater than 2 is expressible as

For example: $46 =$

Example 1. Prove that the integer between a pair of twin primes is always divisible by 3, provided that both twin primes are greater than 3.

Solution. In every three sequential integers, exactly one is:

Let p and $p + 2$ be twin primes and consider the sequential integers

Because the twin primes are assumed greater than 3

Thus:

□

A conjecture is a hypothesis that current evidence suggests could be true, but which has not been proven, so could in fact be false.

While the twin primes conjecture is not yet proven, and not yet contradicted, results in this direction have been established. For example Terence Tao has shown that there must exist some integer N less than 246, so that there are infinitely many primes that are N units apart.

Example 2. Prove that there are infinitely many primes of the form $4n + 3$.

Proof. First we establish a **lemma**, i.e. an

Lemma. Integers of form $4n + 1$ multiply to integers of the same form.

Proof. We multiply two integers of this form

$$(4m + 1)(4n + 1) =$$

□

Assume towards a contradiction there are **finitely** many primes of form $4n + 3$

Consider

$$N =$$

First observe that N has form:

because

$$N =$$

In particular, N is odd, so its factors cannot have form:

By the lemma, it cannot be the case that all the prime factors of N have form

Select a prime factor

Then

and thus:

But this is impossible, so we have arrived at a contradiction.

C. **Dirichlet's Theorem.** We have seen there are infinite primes of form $4n + 3$.

What about of form $4n + 1$? Or, for that matter, of form $7n + 2$?

There is a theorem that unifies these considerations.

Its proof lies beyond the scope of this course. However, we state it nonetheless.

Dirichlet's Theorem. Let $a \perp b$ be relatively prime. There are infinite primes

Example 3. Use Dirichlet's Theorem to prove that there are infinitely many primes whose digits end with 123.

Solution. The following integers are relatively prime.

Therefore the following do not share any prime factors

which means

By Dirichlet's Theorem, there are infinite primes of form

But each such prime ends with digits 123.

□